

Extra Question W3b

Name: _____

1. You are given the following quarterly GDP data from 2010 to 2013:

Time	2010Q1	2010Q2	2010Q3	2010Q4	2011Q1	2011Q2	2011Q3	2011Q4
GDP	123	125	126	125	127	128	130	129

Time	2012Q1	2012Q2	2012Q3	2012Q4	2013Q1	2013Q2	2013Q3	2013Q4
GDP	132	131	133	132	133	134	134	133

- (a) Calculate the 3-quarters Trailing Simple Moving Average (SMA) from 2013Q1 to 2013Q4
- (b) Use the SMA as the trend and calculate short run output (i.e., percentage derivation from the potential output).
- (c)

Sol'n

(a) The SMA at time $t - 1$ is given by:

$$SMA_{t-1} = \frac{Y_{t-3} + Y_{t-2} + Y_{t-1}}{3} = \frac{120 + 130 + 140}{3} = \frac{390}{3} = 130.$$

The SMA at time t is given by:

$$SMA_t = \frac{Y_{t-2} + Y_{t-1} + Y_t}{3} = \frac{130 + 140 + 150}{3} = \frac{420}{3} = 140.$$

(b) The trend estimate is taken as the SMA value, so the estimated trend at time $t - 1$ is 130 and t is 140.

(c) The WMA is computed by:

$$WMA_{t-1} = \frac{w_0 Y_{t-1} + w_{-1} Y_{t-2} + w_{-2} Y_{t-3}}{w_0 + w_{-1} + w_{-2}} = \frac{3(140) + 2(130) + 1(120)}{3 + 2 + 1} = \frac{800}{6} \approx 133.33$$

$$WMA_t = \frac{w_0 Y_t + w_{-1} Y_{t-1} + w_{-2} Y_{t-2}}{w_0 + w_{-1} + w_{-2}} = \frac{3(150) + 2(140) + 1(130)}{3 + 2 + 1} = \frac{860}{6} \approx 143.33$$

(d) The SMA yields 140, whereas the WMA is approximately 143.33. The higher weight on the most recent month (with a sales figure of 150) causes the WMA to be higher.

(e) Because the WMA assigns a larger weight to the most recent observation, it reacts more quickly to changes in recent data. In contrast, the SMA gives equal weight to all observations, smoothing out recent fluctuations.

(f) With weights $w_t = w_{t-1} = w_{t-2} = 1$, the WMA becomes:

$$WMA_t = \frac{1 \cdot Y_t + 1 \cdot Y_{t-1} + 1 \cdot Y_{t-2}}{1 + 1 + 1} = \frac{Y_t + Y_{t-1} + Y_{t-2}}{3},$$

which is exactly the same as the SMA. Therefore, using equal weights does not

emphasize recent observations.